DBMS

Assignment No.- 1

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Title: Design and develop SQL, DDL statements which demonstrate the use of SQL. objects such as ① Table ② View ③ Index ④ Synonym ⑤ different Constraints.

Learning objectives: To learn all type of Data defn lang. Command and their users.

Introduction to SQL

- SQL stands for structured Query Language.
- SQL lets you access and manipulate databases
- SQL is an ANST (American National standards institute) Standard.

1. DDL: DDL is abreuation of Data Definition language it is used to Create and modify the structure of database objects in database.

ex:- CREATE, ALTER, DROP, RENAME, TRUNCATE statement.

1) CREATE TABLE: Create a Table.
2) ALTER TABLE: Modifies a table.
3) Drop TABLE: Delete a table.
4) TRUNCATE TABLE: Delete to all rows in a table

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5) CREATE TABLE : create an index.
6) DROP INDEX : Deletes an index.

10 DDL Commands are:

1. The CREATE TABLE statement.
The CREATE TABLE statement is used to create a table in a database.
Syntax:
CREATE TABLE tablename
(attr 1_name , att 2_name)

2. The DROP TABLE statement
Removes the table from database
Syntax:
DROP TABLE table_name ;

3. The ALTER TABLE statement:
The ALTER TABLE statement is used to add the delete or modify columns in an existing table.
Syntax:
To add a column in a table,
ALTER TABLE table-name.
~~AB~~ ADD Column-name datatype.

4. The RENAME TABLE statement.
Rename the odd table to new table ;

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Syntax.
Rename old-tablename to new-tablename;

5. The TRUNCATE TABLE statement.
The alter table statement is used to truncate a table.
syntax:
To TRUNCATE a table,
TRUNCATE
TABLE table-name ;

6. CREATE VIEW statement
In SQL, a View is a Virtual table based on the result-set of an SQL statement A View Contain rows and column, just like a real table.

Syntax:
CREATE view view-name as
Select Column-name(s)
From table-name
where Condition;

7 SQL Droping a View:
you can delete a view with the DROP VIEW Command
Syntax.
Drop VIEW view-name ?.

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8. CREATE INDES statement:

Syntax:
CREATE INDEX index-name.
on table-name (col_name 2 , col-age 2)
① " index-name is the name of index.
② " table-hume is the name of table to which indexed column belongs.
③ " Column-name 2, Column-name2... is the list of columns which make up the indexes.

9. DROP INDEX Statement
Syntax:
DROP INDEX index_name;

10. CREATE synonyms statements:
Syntax:
CREATE Synonyms Synonname_man For for object-name ?.

Write SQL Quries on the suitable database application using SQL DML Statement.

theory:
DML Command:
Data Manipulation language (DML)

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statements are used for managing data in database. DML commands are not auto - committed.

it means changes made by DML command are not permenant to database it can be rolled back.

1) Insert Command:
Inserts command is used to inserts data into a following table.
Syntax:
Insert into table_name values (data 1, data 2, ...)
eg. s-id, s-name, age.

2) Update Command:
Update Command is used to update table rows.

Syntax:
table_name set Colomn_name = values where condition;
eg. Update student set age = 18 where stud-id = 102;

3) Delete Command.
Delete command is used to delete data from a table

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Syntax:

DELETE from table-name;

eg. DELETE from students;

4. Select Command:

Select is the most important data manipulation on command in SQL. The Select Command Shows the records of specified table.

syntax:

SELECT column_name_1, Column-name_2 ....., Column-name_n.

e.g. select * from student;

SQL Functions:

SQL provides may built-in Functions to perform operations on data. these function are useful while performing mathematical Calculations, String Concatenations, Sub-strings etc. Funn are devided into two Categories.

- Aggregate Functions
- Scalar functions.

- Aggregation Functions

these function returns a single value after calculating from a group of values. Following

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are Some Frequeny used aggregate Functions.

1) AVG ()

- Average returns averages Values after Calculatings From Values in a numeric Column.

Syntax:

SELECT AVG (Column-name) From table name.

e.g. select avg (salary) from Emp;

2) COUNT ()

Count returns the number of rows present in the table either based on some Conditions or without Conditions.

Syntax:

SELECT COUNT (Column-name) From table-name;

3) FIRST ()

First Function returns For Values of Selected Column

Syntax:

SELECT FIRST (Column-name) From table-name.

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4) LAST ()
LAST returns the last Values from Selected Column.

Syntax:
SELECT LAST (Column-name) From table-name.

5) MAX()
MAX Function returns ~~the~~ maximum Value from Selected Column of the table.

Syntax:
SELECT MAX (Column-name) From table-name.

6) MIN ()
MIN Function returns minimum Values from a Selected Column of the table

Syntax:
SELECT MIN (column-name()) From table-name

7) sum ()
sum function returns total sum of a Selected column numeric Values

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Syntax:
SELECT SUM (Column-name) from
table name.

- Scalar functions:

1) UCASE()
UCASE function is used to
Convert values of a string
Column to uppercase character

Syntax:
SELECT UCASE (Column-name).
from table-name.

2) LCASE ()
LCASE functions is used to
Convert values of String
Column to Lowercase character

Syntax:
~~SELEC~~ SELECT LCASE ((olumn-name)
from table-name.

3) MID ()
MID function is used to extract
Substring from Column values
of a string type is in a
table.

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Syntax:
SELECT MID (Column-name, start, length)
from table-name ;

4) ROUND ()
ROUND Function is used to round a numeric field to number OF nearest integer. it is used on decimal point Values
Syntax:

SELECT ROUND (Column_name, decimals)
From table-name.

Conclusion: we have learnt here, DML and DDL Commands and how it use in program.

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